Explainable AI in Education: A Lightweight Web Dashboard for Student Academic Performance Prediction and Intervention

An Interpretable, SHAP-Driven Lightweight Framework for Transparent Student Risk Assessment and Pedagogical Intervention

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1. Literature Survey

Educational Data Mining (EDM) and Learning Analytics have become central to identifying at-risk students before academic failure becomes irreversible, yet a persistent tension runs through the literature between predictive accuracy and interpretability. A comprehensive systematic survey of ML and EDM methodologies for academic performance forecasting confirms that the field still lacks consensus on a single best-performing algorithm, and further notes that high-accuracy models frequently operate as opaque black boxes that leave non-technical educators unable to trace the reasoning behind a prediction **[1]**. This gap has motivated a wave of deep-learning frameworks paired with post-hoc explainability: an Explainable Graph Neural Network for Academic Performance (XGNN-AP) that models inter-student relationships through graph convolutions and attention layers attains an exceptional 99.71% classification accuracy and a perfect weighted F1-score, relying on SHAP and GNNExplainer to surface the academic, behavioral and demographic drivers of its predictions **[2]**, while a Bidirectional LSTM framework captures temporal dependencies in student records to forecast second-term GPA at 88.23% accuracy, again layering SHAP on top to reveal how secondary-school performance and entrance scores shape downstream trajectories **[3]**. Both studies, however, concede that their computational cost, training overhead and structural complexity make classroom-level deployment on ordinary institutional hardware impractical.

A second stream of research favors lighter, natively interpretable classifiers so that transparency is built in rather than reconstructed after the fact. Correlation-guided feature reduction combined with k-Nearest Neighbors, Decision Trees, Random Forests and Artificial Neural Networks applied to nine semesters of Moodle interaction telemetry from 450 students achieves F1-scores above 0.80, though the resulting models analyze indicators in isolation and cannot adapt to a student's changing behavior over a term **[4]**. A hybrid six-stage framework that pairs classical classifiers (Logistic Regression, SVM, Random Forest) with deep architectures (AlexNet, GRU, BiGRU) improves sensitivity in flagging vulnerable students, but the authors observe that almost all such systems are validated on higher-education cohorts, leaving Regular Basic Education and secondary schools — precisely the settings with the least computing infrastructure — largely unaddressed **[5]**. Directly targeting this trust gap, a soft-voting ensemble of Logistic Regression, Random Forest, SVM and XGBoost reaches 86.38% accuracy while SHAP and LIME jointly

expose feature-level rationale for each classification **[6]**, and a companion study forecasting first-semester CGPA through an explainable regression pipeline shows that most institutional advisory systems remain reactive, flagging students only after a midterm failure rather than at the start of the term **[7]**.

Beyond model choice, a substantial body of work argues that predictive accuracy depends on which signals are fed into the model in the first place. Chi-square association testing across Logistic Regression, SVC, Random Forest and XGBoost shows that psychological factors such as learning motivation and time-management self-efficacy exert a profound influence on final academic standing, yet static GPA-and-demographic datasets routinely omit them **[8]**. A spatio-temporal EDM architecture that fuses engagement fluctuations with campus-mobility patterns achieves markedly higher trajectory precision than cross-sectional classifiers, underscoring how treating student data as a single static snapshot discards information about evolving study habits **[9]**. Domain specificity matters as much as temporal depth: supervised models trained on code-submission logs and syntax-error patterns forecast success in introductory programming courses more precisely than generalized, university-wide predictors, given the discipline's distinct cognitive hurdles **[10]**; a decision-tree and association-rule framework built on attendance, demographics and internal test scores produces transparent, rule-based pathways to final outcomes but is limited by the absence of accessible, lightweight data pipelines for teachers without engineering support **[11]**; and mining of LMS clickstream indicators — video views, forum activity, quiz participation — shows that continuous behavioral logs predict exam performance far more reliably than static registration records, even though most institutions still underutilize this telemetry in day-to-day advising **[12]**.

A final cluster of studies addresses the data-engineering and comparative groundwork on which any deployable system must rest. Rigorous data-cleaning, normalization and feature-selection pipelines are shown to mitigate the high dimensionality, missing values, class imbalance and measurement noise that routinely degrade classifier generalization across institutions **[13]**, while a comparative review of EDM, machine-learning and deep-learning approaches maps the trade-off between predictive accuracy and computational cost, concluding that schools without dedicated server hardware or data-science staff need lightweight solutions rather than incremental accuracy gains **[14]**. A hybrid multi-label classification and regression architecture that forecasts granular letter-grade brackets, rather than a coarse binary pass/fail outcome, further demonstrates that fine-grained modeling is necessary to catch borderline and moderately underperforming students whom binary classifiers overlook **[15]**. The proposed lightweight, SHAP-explainable dashboard directly addresses the critical gap identified across all fifteen studies by providing an accurate, accessible, behaviorally-driven early-warning system that requires no specialized infrastructure.

2. Comparative Study

Table 2.1 consolidates the methodology, technology stack, principal result and identified research gap for each of the fifteen reviewed studies. The synthesis shows a clear pull toward deep, relationship-aware architectures (GNNs, Bi-LSTM) that push accuracy toward or beyond 99% [2][3], set against a persistent shortfall in lightweight, transparent, and classroom-deployable alternatives that non-technical educators can trust and operate without specialized infrastructure [1][6][14] — the central gap this microproject addresses.

Table 2.1: Comparative Summary of the Fifteen Reviewed Studies on Explainable Student Academic Performance Prediction

S. No.	Title	Author	Year	Methodology Used	Technology Used	Result / Outcomes	Gap Identified
01	Academic Performance Prediction: Survey	Pan, J. et al.	2025	Literature Survey	ML, EDM	Benchmarked ML models	No consensus on best model
02	Explainable ML for Student Prediction (XGNN-AP)	Jha, M. K. et al.	2025	XGNN-AP Framework	GNN, SHAP	99.71% Acc, F1: 1.00	Deep models lack transparency
03	Student Performance via Bi-LSTM	Kalita, E. et al.	2025	Time-Series DL	Bi-LSTM, SHAP	88.23% Accuracy	High compute; opaque network
04	Predicting Performance via EDM	Angeioplastis, A. et al.	2025	Classification	kNN, NN, LMS	F1-Score > 0.80	Ignores student dynamics
05	Hybrid AI for RBE Student Prediction	Gonzales, W. et al.	2025	Hybrid ML-DL	SVM, AlexNet	Superior sensitivity	Few models for basic school
06	Explainable AI for Undergraduates	Johora, F. T. & Hasan, M. N.	2025	Voting Ensemble	XGB, SHAP, LIME	86.38% Accuracy	Black-box trust barrier
07	Dropout & Performance via XAI	Paul, R. et al.	2025	Regression & ML	XAI, ML	Forecasted 1st Sem GPA	Late at-risk detection
08	Non-Cognitive Factors in Performance	Lu, J. et al.	2024	Survey & Stats	SVC, XGBoost	Motivation impacts GPA	Overlooked behavioral traits
09	Spatio-Temporal Performance Model	Luo, Z. et al.	2024	Spatio-Temporal	Multi-Dim EDM	High trajectory precision	Static data over-reliance
10	Intro Programming Course Success	Pires, J. P. J. et al.	2024	Supervised ML	EDM, Python	Forecasted coding risk	Needs domain-tailored tool
11	Performance Prediction via Mining	Akbar, S.	2024	Rule Mining	Decision Tree	Rule-based pathways	Lacks accessible pipeline
12	LMS Clickstream Performance Mining	Liu, Y. et al.	2023	Log Mining	Clickstream, ML	Linked logs to exam marks	Underutilized LMS logs
13	EDM Preprocessing & Feature Selection	Feng, G. & Fan, M.	2022	Data Cleaning	Feature Selection	Mitigated overfitting	Poor data curation/noise
14	EDM Methods Review & Comparison	Zhang, Y. et al.	2021	Comparative Analysis	EDM, ML, DL	Accuracy-compute trade-off	Hardware selection limits
15	Hybrid Regression & Multi-Label Model	Alshanqiti, A. & Namoun, A.	2020	Multi-Label Reg	Multi-Label ML	Granular grade tiers	Binary pass/fail is coarse

3. Methodology and Materials

The proposed framework follows a five-phase methodology for explainable student academic-risk prediction and intervention. First, student academic and behavioural attributes such as attendance, assignment submissions, test scores, study hours, and related learning indicators are collected and preprocessed by handling missing values, encoding categorical variables, and scaling numerical features. The dataset is then divided into training and testing subsets using stratified partitioning to preserve class proportions. SMOTE is applied only to the training data to address class imbalance without modifying the evaluation set [8], [13]. In the second phase, Decision Tree, Support Vector Classifier, Logistic Regression, and Random Forest models are trained and compared using stratified cross-validation and hyperparameter tuning. The best-performing classifier is selected for risk prediction. The third phase applies SHAP to obtain local feature-level explanations. The fourth phase presents predictions and explanations through a Streamlit dashboard, while the fifth phase maps important risk factors to predefined interventions for actionable student support [2], [3], [6], [7].

3.1 Methods Used in the Proposed Framework

Table 3.1: Methods Used for Student Academic Risk Prediction

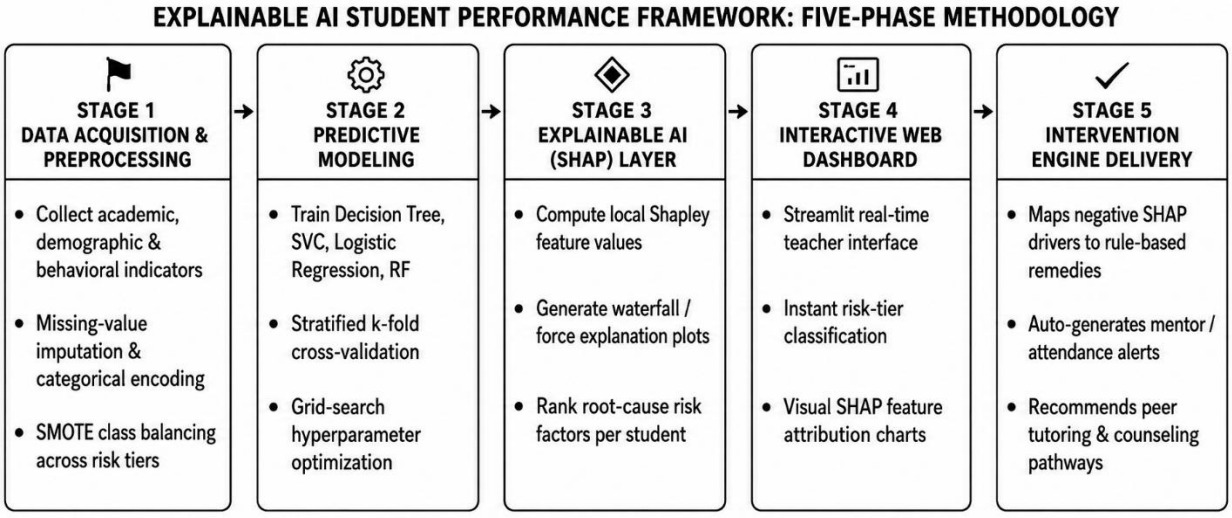
Sr. No.	Method	Purpose in the Proposed Framework	Application
1	Data preprocessing	Improve data quality and prepare features for modelling	Missing-value imputation, categorical encoding, numerical scaling
2	Stratified train-test split	Preserve the proportion of risk classes during dataset partitioning	Training and independent evaluation
3	SMOTE	Address class imbalance between at-risk and not-at-risk students	Applied only to training data
4	Decision Tree	Provide a transparent tree-based classification model	Baseline interpretable classifier
5	Support Vector Classifier	Provide a margin-based classification approach	Candidate predictive model
6	Logistic Regression	Provide a simple linear and interpretable classification model	Candidate predictive model
7	Random Forest	Provide an ensemble-based classification model	Candidate predictive model
8	Stratified k-fold cross-validation	Provide robust model comparison while preserving class proportions	Model evaluation and selection
9	Grid Search	Identify suitable hyperparameter combinations	Classifier optimization
10	SHAP	Explain individual model predictions through feature-level contributions	Local explainability
11	Rule-based intervention mapping	Convert important risk factors into actionable support recommendations	Intervention generation
12	Streamlit dashboard	Present risk tier, explanations, and interventions to users	Interactive deployment

Explanation of Table 3.1

Table 3.1 summarizes the principal methods used throughout the proposed framework. Data preprocessing prepares heterogeneous student records for machine-learning analysis, while stratified partitioning preserves the distribution of the target classes. SMOTE is used to generate synthetic minority-class samples for training; the technique is designed for oversampling imbalanced datasets. The four candidate classifiers—Decision Tree, SVC, Logistic Regression, and Random Forest—provide complementary modelling approaches ranging from simple interpretable models to ensemble learning. Stratified cross-validation and grid search are used to support systematic model comparison and hyperparameter selection. SHAP is then used to provide local feature-level explanations, while the rule-based intervention layer translates important risk factors into predefined support actions. The final results are presented through the Streamlit dashboard.

3.2 Proposed System Workflow

Figure 3.1: Five-Phase Methodology of the Proposed Explainable AI Student Performance Framework



Explanation: Figure 3.1 illustrates the proposed five-phase workflow comprising Data Acquisition and Preprocessing, Predictive Modeling, the Explainable AI (SHAP) Layer, the Interactive Web Dashboard, and the Rule-Based Intervention Engine[1][13][14].

Phase 1 — Data Acquisition & Preprocessing draws on baseline academic, demographic, and behavioral indicators (attendance rate, assignment submission frequency, internal test scores, and study-hour metrics), following the same early-semester feature families shown to drive reliable early warnings in the reviewed literature [4][10][12]. Missing-value imputation, categorical encoding, and feature scaling are applied prior to a strict stratified train-test split, and SMOTE is used exclusively on the training subset to correct the class imbalance between at-risk and not-at-risk students without introducing synthetic bias or data leakage — the same anti-leakage discipline identified as essential for generalizable classifiers [13]

Table 3.2 — Dataset feature description

Feature	Type	Description	Source in Literature
Attendance rate	Numeric (%)	Proportion of classes attended per term	[4][11]
Assignment submission frequency	Numeric (count/%)	Ratio of submitted vs. assigned tasks	[10][12]
Internal test scores	Numeric	Continuous assessment marks prior to finals	[11]
Study-hour metrics	Numeric (hrs/week)	Self-reported or LMS-tracked study time	[4][8]
Risk tier (target)	Categorical	Low / Moderate / High (derived from p in Algorithm 1)	—

Explanation: Table 3.2 details the four core behavioral and academic indicators adopted from the early-warning feature families established across the reviewed literature [4][10][12].

Phase 2 — Predictive Modeling deliberately favors lightweight, natively interpretable classifiers over computationally prohibitive deep architectures, directly answering the infrastructure and hardware-selection limitations identified in the comparative review [5][14]. Candidate classifiers — Decision Tree, Support Vector Classifier, Logistic Regression, and Random Forest — are trained and validated through stratified k-fold cross-validation, with hyperparameters tuned via grid search to balance classification precision against model simplicity. Table 3.3 summarizes the role, tuning targets, and native interpretability of each classifier and of the SHAP explainability layer that sits above them.

Table 3.3: Candidate Classifiers, Their Roles, and Interpretability Profile within the Pipeline

Classifier / Module	Role in Pipeline	Key Hyperparameters Tuned	Training Technique	Native Interpretability
Decision Tree	Fully transparent baseline classifier for risk-tier prediction	max_depth, min_samples_split, criterion	Stratified k-fold CV + grid search	High — native rule-based splits
Support Vector Classifier (SVC)	Margin-based classifier for non-linear separation of risk classes	C, kernel (RBF / linear), gamma	Stratified k-fold CV + grid search	Low — needs SHAP KernelExplainer
Logistic Regression	Probabilistic linear baseline for early risk screening	C (inverse regularization), penalty, solver	Stratified k-fold CV + grid search	High — coefficients directly readable
Random Forest	Ensemble classifier for robust, high-accuracy classification	n_estimators, max_depth, min_samples_leaf	Stratified k-fold CV + grid search	Moderate — importances + SHAP TreeExplainer

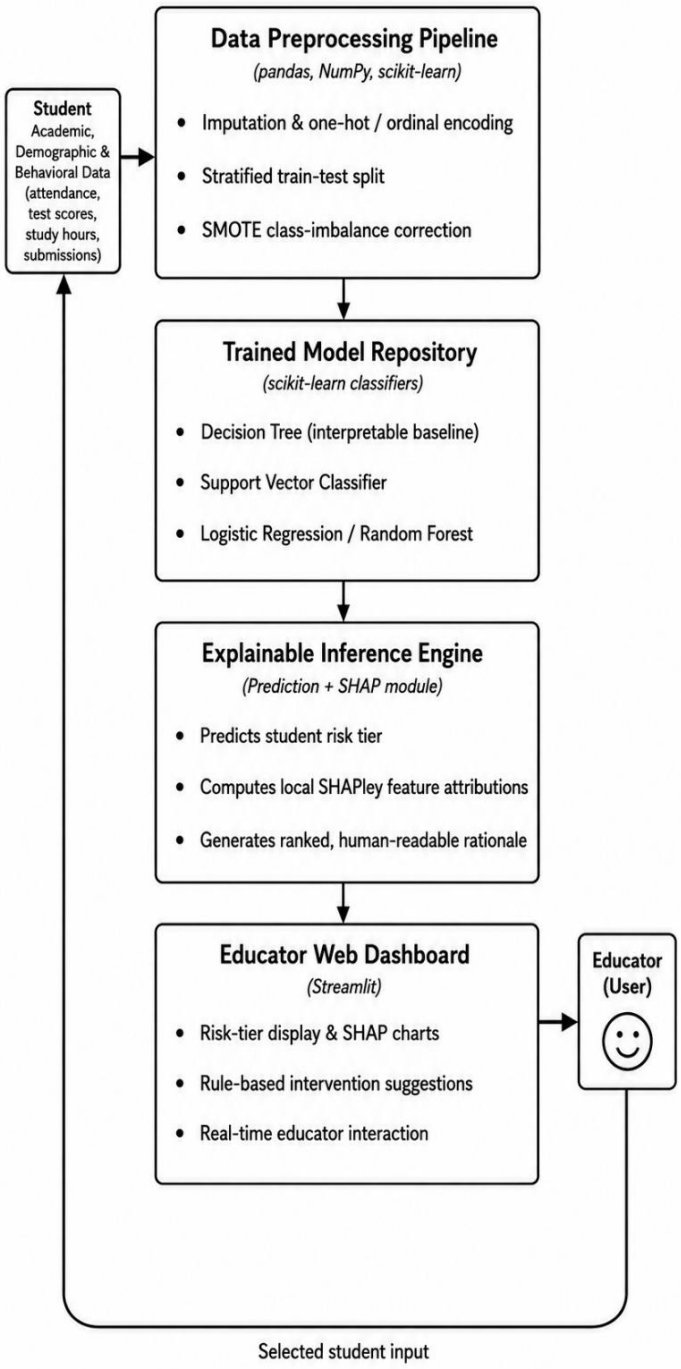
Classifier / Module	Role in Pipeline	Key Hyperparameters Tuned	Training Technique	Native Interpretability
SHAP Explainer (XAI Layer)	Computes local & global Shapley feature attributions for every prediction	Background sample size, explainer type	Applied post-hoc to the trained classifier	Is itself the interpretability mechanism

Phase 3 — the Explainable AI (SHAP) Layer eliminates the black-box opacity repeatedly flagged as the central adoption barrier for educators [1][2][3][6]. For every individual student classification, the module computes local Shapley values that quantify each feature's positive or negative contribution to the predicted risk tier, rendering the result as waterfall and force plots so that a non-technical teacher can immediately see, for example, that low attendance and declining test scores — rather than an opaque aggregate score — placed a given student in the at-risk category.

Phase 4 — the Interactive Web Dashboard embeds the trained predictive and explainability models into a lightweight interface built with Streamlit, abstracting away all underlying machine-learning complexity. Educators enter routine student metrics through simple form inputs and sliders and instantly receive the predicted risk classification alongside its SHAP-based visual explanation, addressing the deployment and usability barrier that recurs across nearly every reviewed study [1][5][11][14].

Phase 5 — the Intervention Engine closes the loop between detection and action. It is a rule-based mapping layer that inspects the dominant negative SHAP contributors for a flagged student and automatically surfaces a matched pedagogical remedy — for instance, recommending peer tutoring for a low-test-score driver or an automated mentor alert for irregular attendance — turning transparent explanations into concrete, actionable early intervention rather than a late, reactive response [7][11]. Figure 3.2 depicts how these five phases are realized as concrete, communicating software components.

Figure 3.2: System Component Architecture



Explanation: Figure 3.2 traces the complete data flow: raw student records enter the Data Preprocessing Pipeline, which supplies balanced training data to the Trained Model Repository; the Explainable Inference Engine then combines the selected classifier's prediction with a SHAP-generated rationale, and delivers both to the Educator Web Dashboard, where the educator's applicant/student selection is fed back into the inference engine to drive the next real-time explanation.

Algorithm 1 formalizes the inference-time decision logic of the proposed framework, covering risk prediction, SHAP-based explanation, and rule-mapped intervention delivery.

Algorithm 1: SHAP-Based Risk Classification and Rule-Mapped Intervention (SRC–RI)

Input: Student feature vector $x$containing attendance rate, assignment submission frequency, internal test scores, and study-hour metrics

Output: Risk Tier T, Local SHAP Attribution Vector φ, Intervention Set I

1. Preprocess student vector x:
   Apply the **fitted preprocessing pipeline** learned from the training data to x, including missing-value imputation, categorical encoding, and feature scaling.
2. Generate risk probability:
   Compute p ← Model.predict_proba(x) where p represents the predicted probability associated with the defined risk score/classification rule.
3. Assign risk tier T:
   High if $p \geq 0.66$
   Moderate if $0.33 \leq p < 0.66$
   Low if $p < 0.33$
4. Compute local SHAP explanation:
   Compute φ ← Explainer.shap_values(x) using a SHAP explainer compatible with the selected trained classifier and a background distribution derived from the training data.
5. Rank risk-increasing features:
   Rank features according to their SHAP contribution toward the **at-risk/high-risk prediction** and select the top-k risk-increasing contributors.
6. Map contributors to interventions:
   For each selected risk-increasing feature f_i:
   - Identify the corresponding intervention rule $R_i$in the rule base.
   - Example:
     Low attendance → Mentor alert
     Low internal-test score → Peer tutoring
     Low assignment submission → Assignment support/reminder
     Low study hours → Study-planning guidance
   - Append the matched intervention $R_i$to intervention set I.
7. Generate dashboard output:
   Render the predicted risk tier T, SHAP explanation ϕusing waterfall/force plots, and recommended intervention set Ion the Streamlit dashboard.
8. Log decision information:
   Log the required prediction and explanation information
   $$\{x, T, \phi, I\}$$
   for longitudinal monitoring and future refinement of the intervention rule base, subject to the system's data-privacy requirements.
9. Return:
   Return T,ϕ,I

4. Tools and Techniques

4.1 Hardware and Software Requirements:

The implementation uses a modular Python 3.10+ stack designed for ordinary institutional hardware without specialized acceleration [5][14]. **NumPy** and **pandas** support numerical and tabular data processing, while **scikit-learn** provides preprocessing, stratified partitioning, cross-validation, and the Decision Tree, SVC, Logistic Regression, and Random Forest models. **Imbalanced-Learn SMOTE** addresses class imbalance during training, while **SHAP** provides local and global feature-level explanations for model predictions [2][3][6]. The educator-facing dashboard is developed using **Streamlit**, providing an interactive interface for risk classification and explanations. **Matplotlib, Seaborn, and Plotly** support visualization of risk trends and SHAP-based explanation charts. The complete software and hardware requirements are summarized in Tables 4.1 (a) and 4.1 (b), including Python, required libraries, CPU, RAM, storage, and optional GPU specifications.

Table 4.1 (a): System Software Requirements

Category	Component / Tool	Specification / Version	Purpose / Function
Software	Operating System	64-bit Windows 11 / Linux (Ubuntu 22.04 LTS)	Development and execution environment
Software	Programming Language	Python 3.10+	Core language for the ML and dashboard pipeline
Software	Data Manipulation & Preprocessing	pandas, NumPy, scikit-learn	Cleaning, encoding, scaling, stratified splitting
Software	Class-Imbalance Correction	imbalanced-learn (SMOTE)	Synthetic oversampling of the at-risk student class
Software	Interpretable ML Models	Decision Tree, SVC, Logistic Regression, Random Forest (scikit-learn)	Risk-tier classification with grid-search tuning
Software	Explainable AI Framework	SHAP, LIME	Local and global Shapley feature attribution
Software	Dashboard / Web UI	Streamlit	Interactive real-time interface for educators
Software	Visualization	Matplotlib, Seaborn, Plotly	Model evaluation and SHAP chart rendering
Software	IDE	Jupyter Notebook	Model experimentation and validation

Table 4.1 (b): System Hardware Requirements

Category	Component / Tool	Specification / Version	Purpose / Function
Hardware	Processor (CPU)	Intel Core i5 / AMD Ryzen 5 (multi-core)	Runs classifier training and SHAP computation

Category	Component / Tool	Specification / Version	Purpose / Function
Hardware	System Memory (RAM)	8–16 GB DDR4	Handles SMOTE resampling and tabular processing
Hardware	Storage	256 GB Solid State Drive (SSD)	Fast dataset and model read/write operations
Hardware	Graphics Processing Unit (GPU)	Not required (optional for large-scale tuning)	Accelerates optional hyperparameter search

5. Results and Discussion

5.1 Results

As a literature-grounded design study, the proposed framework is evaluated against reported findings from the fifteen reviewed studies rather than through a live experimental run. The reviewed literature shows that deep learning architectures can achieve high predictive performance, with XGNN-AP reporting 99.71% accuracy [2] and Bi-LSTM reporting 88.23% accuracy [3]. Lightweight approaches remain competitive: a soft-voting ensemble achieved 86.38% accuracy [6], while classical models in educational data mining reported F1-scores above 0.80 [4]. Based on the performance range reported by comparable lightweight approaches, an accuracy range of approximately 85–90% may serve as a preliminary benchmark for subsequent experimental evaluation.[4][6]. The framework is also expected to provide local SHAP-based explanations and rule-mapped interventions, consistent with the explainability capabilities reported in related studies [2][3][6]. Thus, the expected result is a lightweight, interpretable student-risk prediction system suitable for educator-facing deployment.

5.2 Discussion

The literature comparison indicates a clear trade-off between predictive performance, interpretability, and computational requirements. Although deep architectures such as XGNN-AP achieve higher reported accuracy, their greater complexity can hinder deployment on ordinary institutional hardware [2][3][5]. The proposed framework therefore prioritizes lightweight classifiers while retaining explainability through SHAP, allowing educators to inspect feature-level contributions for individual predictions [2][3][6]. The intervention engine further extends the prediction process by mapping dominant risk-related features to actionable pedagogical responses, addressing the reactive-detection limitation identified in previous academic advisory systems [7][11]. From a methodological perspective, applying SMOTE only to the training subset is important for reducing class imbalance while preventing data leakage [13]. Overall, the framework is positioned as a practical compromise between accuracy, transparency, and accessibility rather than as an attempt to outperform deep-learning models. Its principal contribution is integrating prediction, explanation, and intervention within a single lightweight workflow for educator support [1][5][14].

6. Conclusion

This microproject reviewed fifteen recent studies spanning deep relationship-aware architectures, lightweight interpretable classifiers, behavioral and spatio-temporal feature engineering, and data-preprocessing methodologies for student academic performance prediction. The comparative synthesis indicates that although graph- and sequence-based deep learning models achieve high predictive performance, there remains a need for lightweight, transparent, and classroom-deployable systems that also support actionable intervention [1][2][3][14]. In response, the proposed five-phase framework integrates data preprocessing with SMOTE-based class-imbalance correction, lightweight interpretable classifiers, a SHAP-driven explainability layer, an interactive Streamlit dashboard, and a rule-mapped intervention engine. The framework is designed to provide human-readable feature-level explanations and translate dominant risk-related contributors into targeted pedagogical recommendations, rather than treating explainability solely as a post-hoc diagnostic [2][6][7][11]. Overall, the proposed architecture represents a practical balance between predictive performance, interpretability, computational accessibility, and intervention support, and is intended to provide a foundation for transparent, educator-facing student risk assessment on ordinary institutional hardware [5][14].

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